1. AIM

To simulate and study the Replacement Gate (Gate-Last) CMOS fabrication process and virtual metrology using Coventor SEMulator3D. To evaluate aluminium diffusion profiles, investigate the sensitivity of gate critical dimension (CD) to gate etch parameters, characterize void formation failure modes under etch ratio variations, and assess device functionality under lithography bias and overlay misalignment.

2. TOOL USED

Coventor SEMulator3D (V2_ReplacementGate.zam, RMG_metrology.vanalysis)

3. THEORY

Replacement Metal Gate (RMG) Process Architecture:

The Replacement Gate (RMG) process, commonly referred to as the Gate-Last scheme, is an advanced CMOS integration approach introduced at sub-45 nm nodes to overcome fundamental limitations of traditional Gate-First polysilicon flows. In Gate-Last integration, a sacrificial (dummy) polysilicon gate is fabricated first to define the channel geometry and withstand high-temperature source/drain implant activation anneals. Subsequently, dielectric planarization is performed, the dummy gate is selectively etched away, and high-k dielectric and replacement metal gate stacks (e.g., TiN, Al, W) are deposited into the narrow cavity at lower thermal budgets.

Refractory Dummy Gate & Aluminium Thermal Diffusion:

Polysilicon is refractory and prevents gate material migration during the heavy source/drain thermal budget. In contrast, low-resistance gate metals such as aluminium (Al) are vulnerable to rapid interstitial diffusion into the channel at elevated temperatures. Excessive metallic contamination in the active channel degrades carrier mobility, increases junction leakage, and shifts threshold voltage. A dummy anneal step models the cumulative thermal budget, and the device fails if aluminium contamination exceeds 10^{12} cm^{-3} in the active region.

Virtual Metrology and Gate Critical Dimension (CD):

Virtual metrology utilizes algorithmic 3D surface tracking to extract critical geometric dimensions across arbitrary cross-sections. Gate Critical Dimension (CD) represents the physical gate length (L_g) and is acutely sensitive to lithography bias, etch depth, and chemical etch selectivity (ratios) between silicon, silicon dioxide (SiO2), and silicon nitride (Si3N4).

Void Formation Mechanism:

Depositing conformal high-k dielectrics and multi-layer metal liners into extreme aspect-ratio trenches poses severe step-coverage challenges. If etch parameters cause the trench opening to narrow or if lateral oxide undercut creates re-entrant profiles, metal accumulates at the trench neck, pinching off the cavity before complete bottom-up fill is achieved, producing an enclosed void.

4. FABRICATION RECIPE (PROCESS EDITOR STEPS)

Table 4: Fabrication Recipe (Process Editor Steps)

Step	Action	Step Name	Material(s)	Depth (nm)	Mask
1	Wafer Setup	Wafer Setup	Crystalline Silicon	100	-
2	Track Void	Enable Void Tracking	-	-	-
3.1	Conformal Deposit	Gate oxide deposition	SiO2 (Thermal)	5	-
3.2	Conformal Deposit	Gate polysilicon deposition	Amorphous Silicon	40	-
3.3	Smooth Planarize	SiARC Deposition	Si3N4	20	-
3.4	Planarizing Deposit	Dummy gate resist coating	Organics / Resist	50	-
3.5	Expose Material	Expose resist for gate	Organics / Resist	0	Gate pattern
3.6	Multi Etch	Multi Etch (Regular)	Nitrides / Oxides / Silicon	70	-
3.7	Remove Materials	Remove organics	Organics / Resist	-	-
4.1	Conformal Deposit	Conformal deposition	Si3N4	5	-
4.2	Planarizing Deposit	Source drain patterning	Organics / Resist	30	-
4.3	Expose Material	Expose resist for source drain	Organics / Resist	0	Source Drain
4.4	Enhanced Basic Etch	Basic Etch	Nitrides / Organics	30	-
4.5	Implant	Implant	N-Type / P-Type	-	-
4.6	Remove Materials	Remove resist	Organics / Resist	-	-
4.7	Planarizing Deposit	Conformal deposition	SiO2 (HDP)	5	-
4.8	Polish	Stopping CMP	-	-	-
5	Enhanced Basic Etch	Dummy gate cap etch	Si3N4	20	-
6	Enhanced Basic Etch	Dummy gate silicon etch	Silicon / Oxides	50	-
7	Conformal Deposit	Gate high-k deposition	HfOx	2	-
8	Visibility Deposit	Aluminium deposition	Aluminium	2	-
9	Visibility Deposit	TiN deposition	TiN	2	-
10	Planarizing Deposit	Filling gate metal	W	5	-
11	Polish	Stopping CMP	-	-	-

Step	Action	Step Name	Material(s)	Depth (nm)	Mask
12	Diffusion	Dummy Anneal	Aluminium / Other materials	-	-

5. REPLACEMENT GATE FABRICATION RESULT

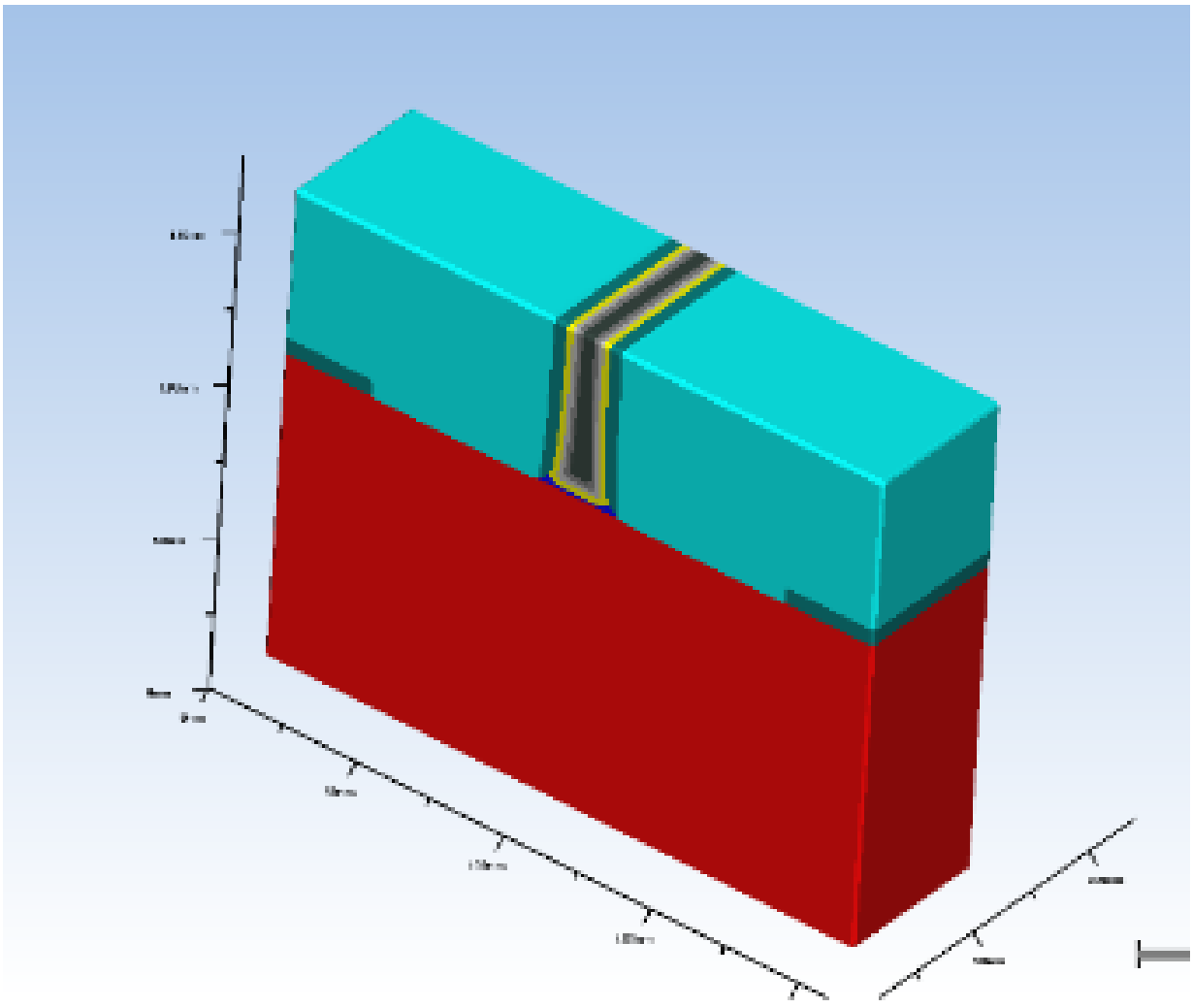


Figure 1: Final 3D Structure of Replacement Metal Gate (RMG) Process

6. QUESTION 1: ALUMINIUM DIFFUSION ANALYSIS

Question 1(a): Aluminium Concentration Profile After Dummy Annealing

Problem Statement: The polysilicon dummy gate is refractory and does not diffuse. Aluminium diffusion occurs during subsequent high-temperature thermal cycles. A dummy anneal was incorporated at the end of the process to represent the combined effect of all anneals. Plot the metal diffusion in the structures after the dummy anneal. What is the Al concentration in the active region? Use the 'Aluminium Profile Export' analysis result to compute the average concentration in the active region. Is the device working?

Experimental Methodology & Results:

The aluminium diffusion concentration field was analyzed along two distinct cross-sectional paths through the device:

1. Sloped Path (Simultaneous coordinate change along X and Z axes across the gate dielectric-silicon channel interface):

• Minimum Concentration = 0 cm^{-3}

• Maximum Concentration = 18,782.6 cm^{-3}

• Average Concentration = 2,074.97 cm^{-3}

2. Horizontal Path (Uniaxial variation along the X coordinate directly beneath the gate in the active channel):

• Minimum Concentration = 2.15016 x 10^{5} cm^{-3}

• Maximum Concentration = 1.38574 x 10^{7} cm^{-3}

• Average Concentration = 8.29102 x 10^{6} cm^{-3}

Evaluation of Device Operation:

The maximum permissible aluminium contamination threshold specified for reliable transistor operation is:

$C_limit = 10^{12}$ cm^{-3}

Comparing the extracted values to the limit:

• Sloped path average: 2,074.97 cm^{-3} << 10^{12} cm^{-3} (margin > 8 orders of magnitude)

• Horizontal path average: 8.29102 x 10^{6} cm^{-3} << 10^{12} cm^{-3} (margin > 5 orders of magnitude)

Both average concentration values across the active channel region are substantially below the 10^{12} cm^{-3} failure limit. Therefore, the device satisfies the contamination criterion and is fully working.

7. ALUMINIUM CONCENTRATION FIELD

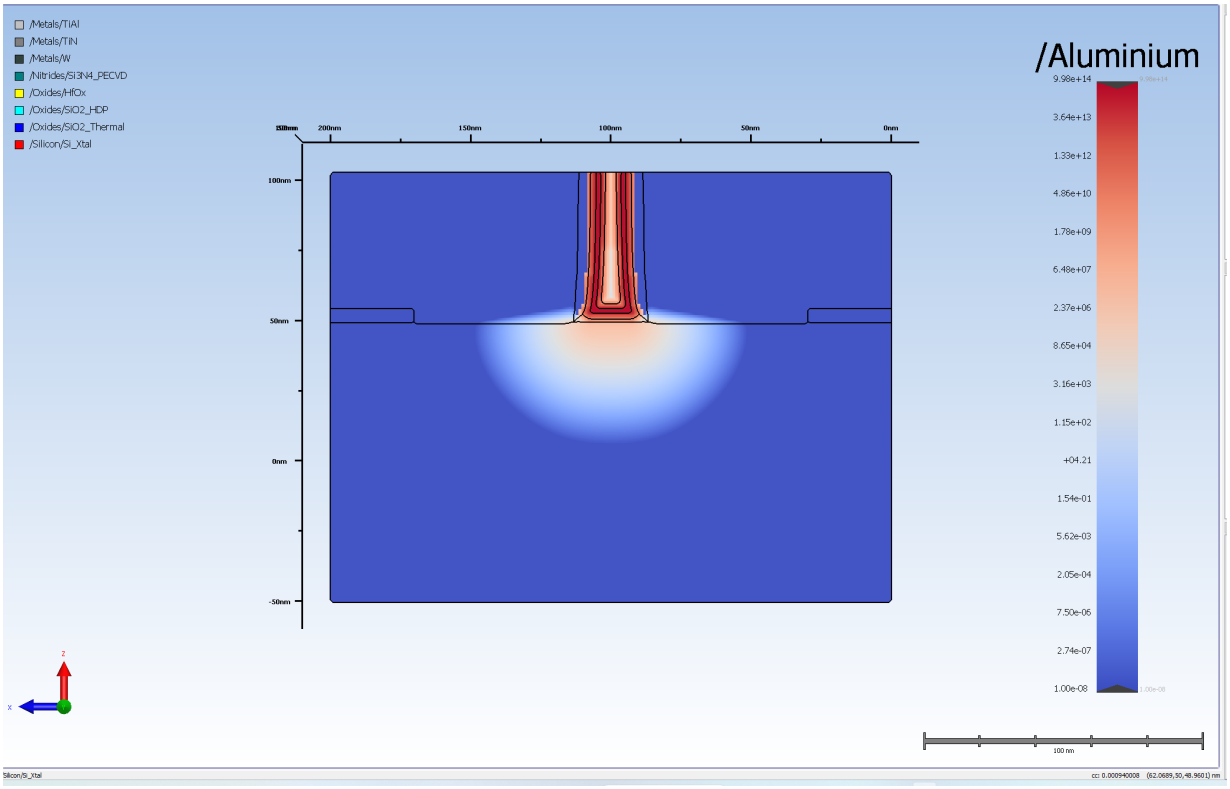


Figure 2: Aluminium Concentration Field Color Map After Dummy Annealing

8. QUESTION 2: METROLOGY OF REPLACEMENT GATE PROCESS

(a) Key Process Parameters Affecting the Replacement Gate Process:

The RMG process is most strongly influenced by five critical fabrication parameters:

1. Lithography Bias: Directly defines the dummy gate trench width. Any deviation in exposure sizing shifts the final physical gate length (Gate CD).

2. Gate Etch Parameters: Etch depth, etch duration, and chemical selectivity (etch ratios between SiO2, Si3N4, and Si) control sidewall slope, cavity taper, and bottom CD.

3. Metal Deposition Parameters: Film thickness, aspect-ratio coverage, and deposition conformality for high-k (HfOx), TiN, Al, and W fill dictate cavity filling integrity and void avoidance.

4. Overlay Alignment: Lateral mask registration accuracy between gate patterning and source/drain diffusion regions prevents asymmetric overlap capacitances and punch-through.

5. Annealing Conditions: Temperature, dwell time, and ambient environment govern dopant activation versus harmful metallic inter-diffusion.

(b) Virtual Metrology: Gate CD vs. Gate Etch Parameters:

Using the SEMulator3D virtual metrology module (RMG_metrology.vanalysis), the quantitative sensitivity of Gate Critical Dimension (CD) to gate etch depth and etch selectivity ratios was characterized:

• Gate CD vs. Etch Depth: As gate etch depth increases, Gate CD exhibits a monotonic decrease:

- Etch Depth = 60 nm -> Gate CD = 19.15 nm

- Etch Depth = 70 nm -> Gate CD = 13.56 nm

- Etch Depth = 80 nm -> Gate CD = 11.43 nm

• Gate CD vs. Etch Ratios: Chemical etch selectivity ratios for SiO2 and Nitride dramatically modulate sidewall profile and lateral trimming.

(c) Two Examples to Decrease Gate CD:

Based on the virtual metrology response curves, two distinct methods can be employed to achieve smaller Gate CD:

1. Increase Gate Etch Depth: Deepening the vertical etch from 60 nm to 80 nm removes additional gate profile material, decreasing Gate CD from 19.15 nm to 11.43 nm (a 40.3% reduction).

2. Increase Nitride Etch Ratio: Increasing the nitride etch ratio from 0.75 to 2.0 provides greater lateral clearance of nitride spacers, reducing Gate CD from 16.92 nm to 6.51 nm (a 61.5% reduction).

9. GATE CD VIRTUAL METROLOGY RESPONSE

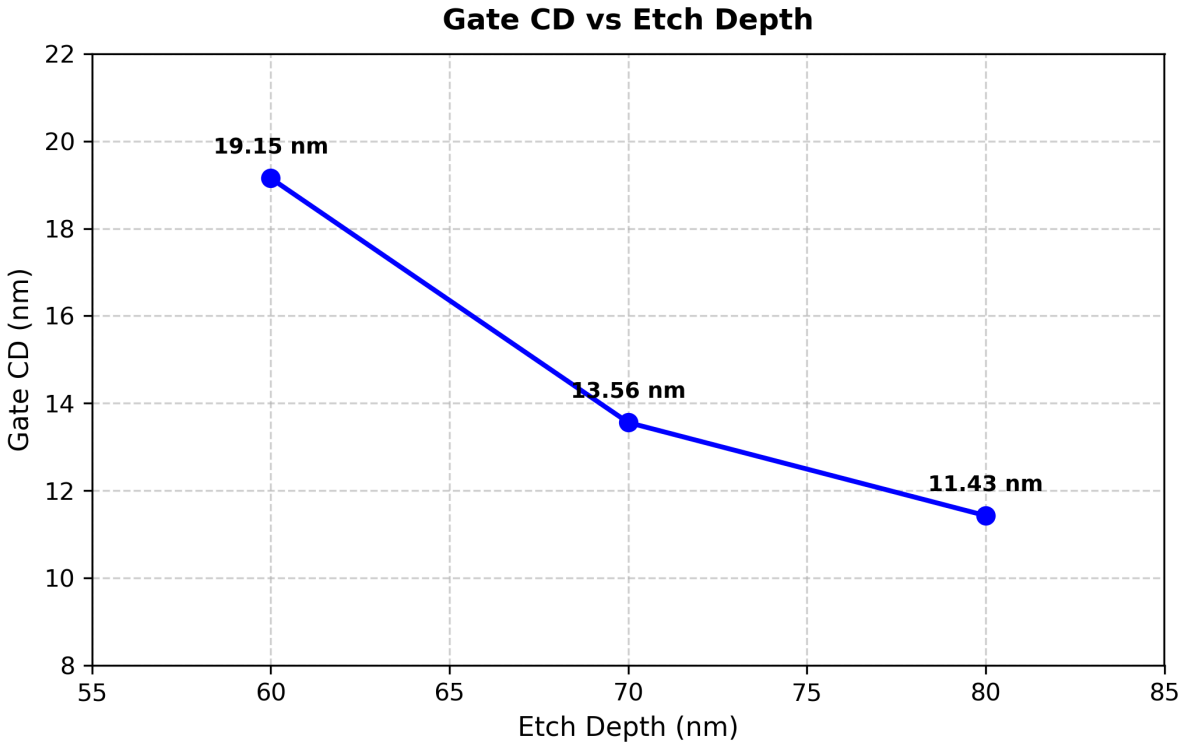


Figure 3: Gate Critical Dimension (CD) vs. Gate Etch Depth

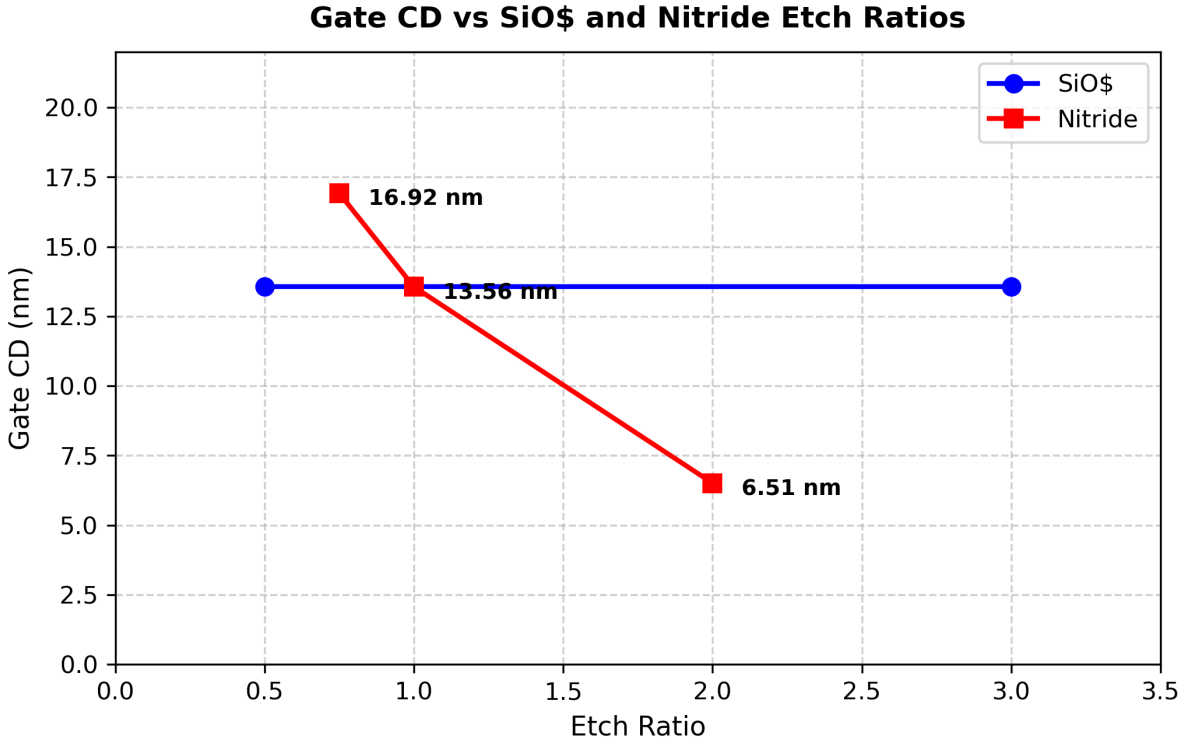


Figure 4: Gate Critical Dimension (CD) vs. SiO2 and Nitride Etch Ratios

10. QUESTION 2 (CONTINUED): LITHOGRAPHY BIAS, VOIDING & OVERLAY

(d) Effect of 5 nm Lithography Bias on Device Operation:

A 5 nm lithography bias was applied at the gate lithography step, and the RMG flow was executed. Extracted aluminium diffusion concentrations along both calibration paths are:

• Sloped Line: Minimum = 0 cm^{-3}, Maximum = 9.72999 x 10^{-5} cm^{-3}, Average = 8.17491 x 10^{-6} cm^{-3}

• Horizontal Line: Minimum = 7.61023 x 10^{-5} cm^{-3}, Maximum = 1.74253 cm^{-3}, Average = 0.307131 cm^{-3}

Result: Both average concentrations remain negligible and substantially below C_limit = 10^{12} cm^{-3}. The device remains fully functional.

(e) Failure Mode Analysis: Void Formation Under Etch / Deposition Ratio Mismatch:

Returning to the baseline recipe and altering the SiO2 etch ratio from 0.5 to 2.0 causes excessive lateral undercut in the oxide liner, forming an overhanging cavity profile. During subsequent tungsten (W) metal deposition, metal builds up preferentially near the top constriction. Pinch-off occurs before the lower volume fills, creating an enclosed void defect inside the replacement gate.

(f) Void Metrology Analysis:

Using the SEMulator3D 'Void metrology' feature, the physical dimensions of the trapped cavity void were extracted:

• Length of Void: 24.5 nm

• Maximum Width of Void: 4.6 nm

• Cross-Sectional Area of Void: 54.6 nm^2

(g) Effect of Source_Drain2 Mask & 10 nm X-Overlay Alignment Error:

The source/drain lithography mask was replaced with 'Source_Drain2', and an intentional 10 nm overlay error in the X-direction was introduced at gate lithography. The resulting aluminium profiles:

• Sloped Line: Minimum = 0 cm^{-3}, Maximum = 18,782.6 cm^{-3}, Average = 2,074.97 cm^{-3}

• Horizontal Line: Minimum = 2.15016 x 10^5 cm^{-3}, Maximum = 1.38574 x 10^7 cm^{-3}, Average = 8.29102 x 10^6 cm^{-3}

Result: Active region aluminium concentration remains far below C_limit = 10^{12} cm^{-3}, confirming the transistor withstands 10 nm lateral overlay error without electrical compromise.

11. ALUMINIUM PROFILE: 5 NM LITHO BIAS

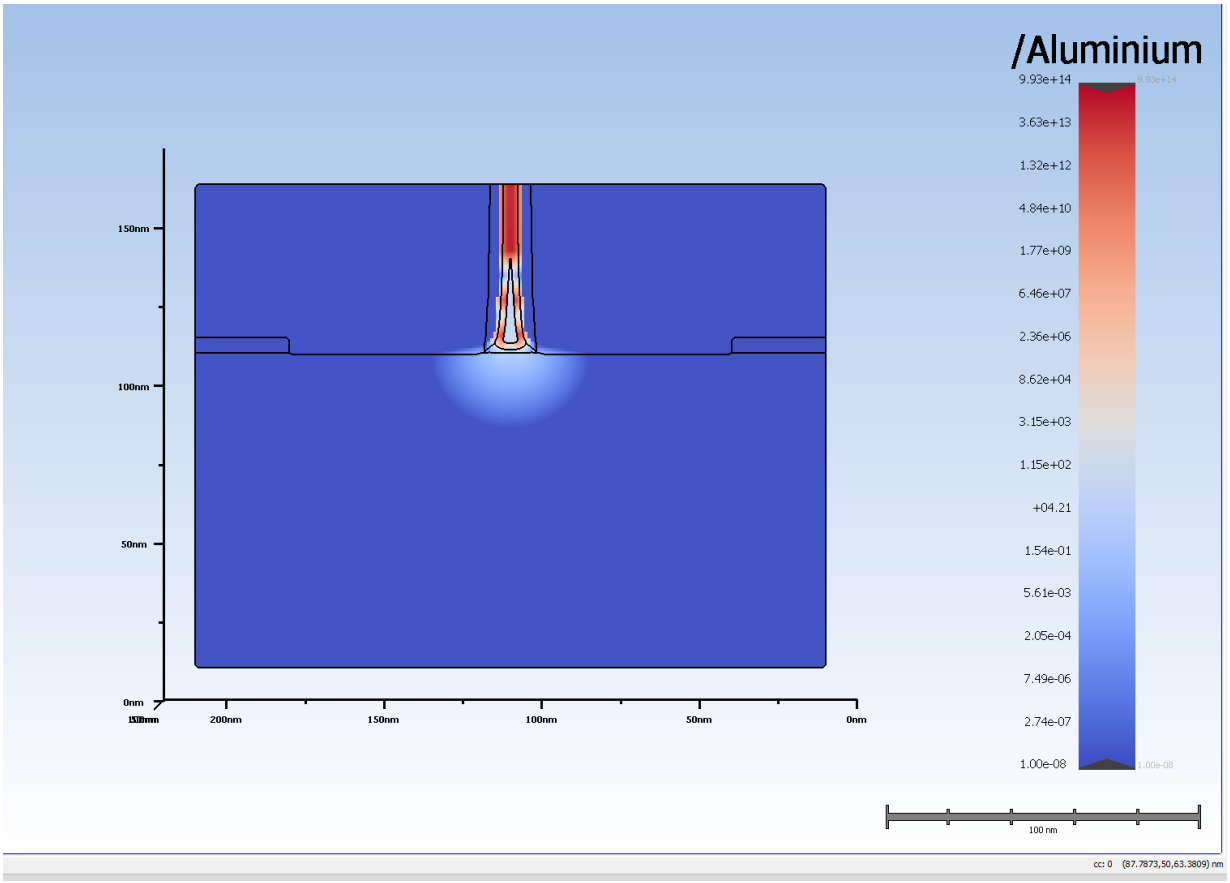


Figure 5: Aluminium Concentration Field After 5 nm Lithography Bias

12. GATE CAVITY VOID DEFECT ANALYSIS

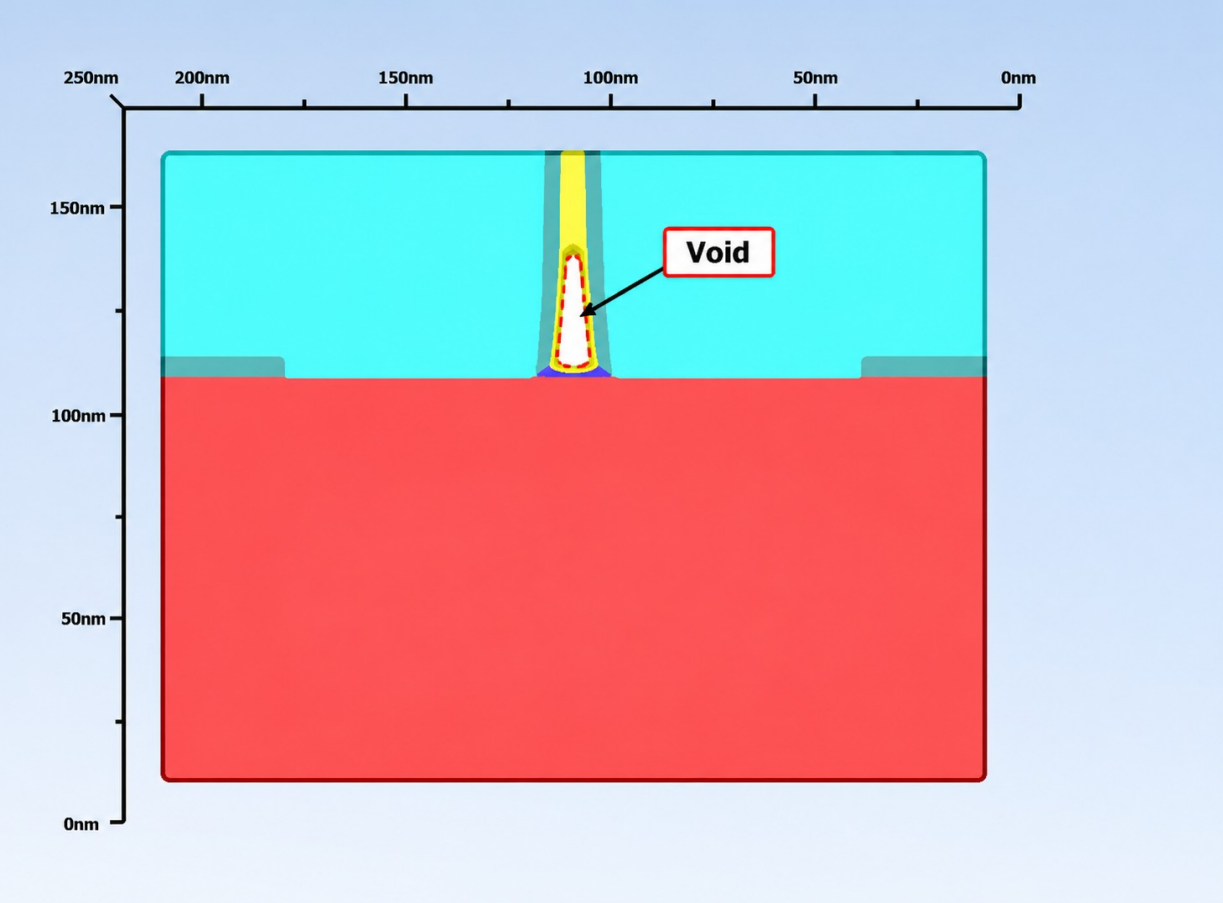


Figure 6: Void Defect Formation Inside Gate Cavity (SiO2 Etch Ratio = 2)

13. ALUMINIUM PROFILE: 10 NM X-OVERLAY

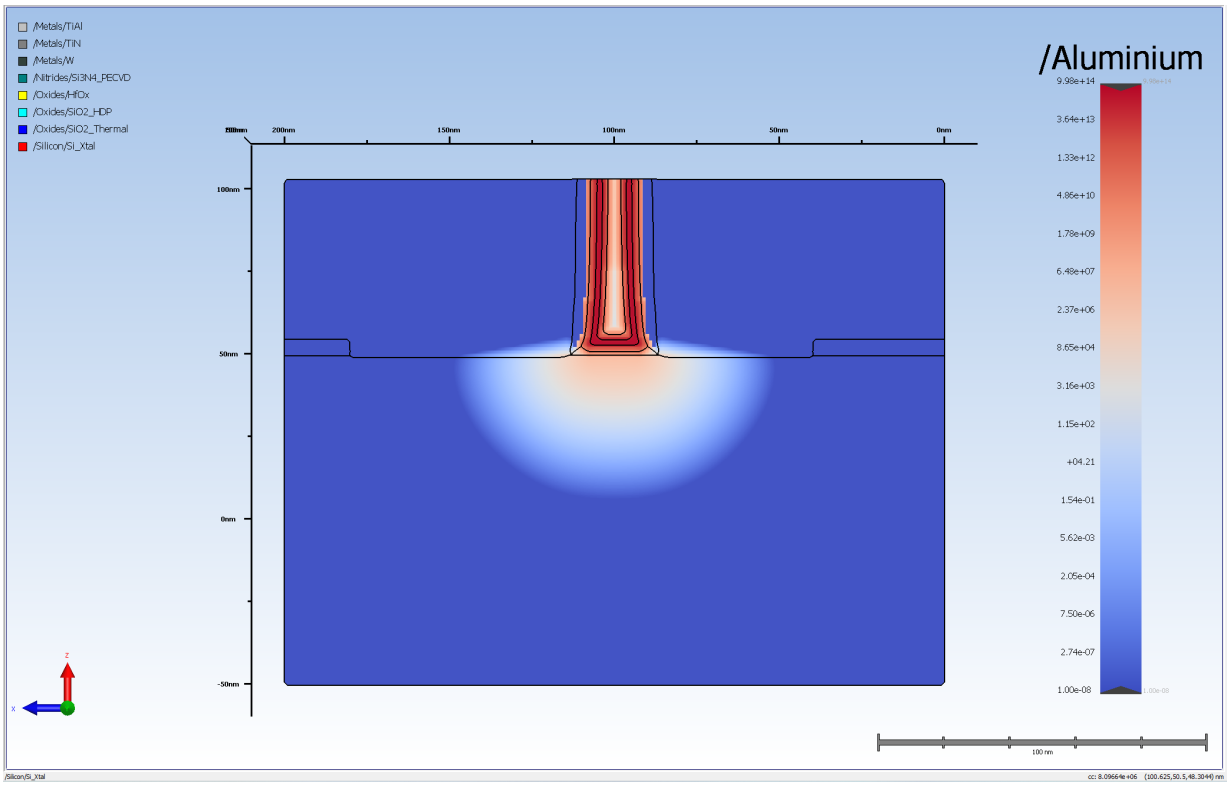


Figure 7: Aluminium Concentration Field After Source_Drain2 Mask & 10 nm X-Overlay

14. OBSERVATIONS & CONCLUSION

Observations:

1. In the baseline Replacement Metal Gate (RMG) flow, average aluminium diffusion in the active channel is 2,074.97 cm^{-3} (sloped path) and 8.29102 x 10^{6} cm^{-3} (horizontal path), well below the 10^{12} cm^{-3} contamination threshold.

2. Virtual metrology reveals that Gate CD scales inversely with gate etch depth (60 nm -> 19.15 nm; 80 nm -> 11.43 nm) and nitride etch ratio (0.75 -> 16.92 nm; 2.0 -> 6.51 nm).

3. Increasing the SiO2 etch ratio from 0.5 to 2.0 induces premature neck pinch-off during tungsten metallization, trapping a void of length 24.5 nm, width 4.6 nm, and area 54.6 nm^{2}.

4. The RMG flow exhibits robust design margin: neither a 5 nm lithography bias nor a 10 nm X-direction overlay error drives channel aluminium contamination above the 10^{12} cm^{-3} threshold.

Conclusion:

The Replacement Gate (Gate-Last) process flow and virtual metrology were comprehensively investigated using Coventor SEMulator3D. The thermal isolation provided by the sacrificial polysilicon dummy gate ensures minimal metal inter-diffusion into the active region during source/drain thermal anneals. Process sensitivity analysis demonstrated precise control levers for Gate CD scaling, while void metrology highlighted the strict aspect-ratio and etch selectivity constraints required for defect-free gate filling. The device structure demonstrated high structural and thermal integrity across all tolerance variations.